

Exercises 09/09

1. Recall that two vectors in $\mathbb{R}^n$ are orthogonal if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$. Let $\mathbf{v} \in \mathbb{R}^n$. Interpreted as a linear functional, what does the kernel of $\mathbf{v}$ look like? As an explicit example, let $n = 2$ and $\mathbf{v} = (1, 2)$.
2. Show that the map that takes a list of numbers to its mean is a linear functional.
3. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be given by

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \mapsto \begin{pmatrix} x_1 - 4x_3 \\ x_2 \end{pmatrix}$$

Determine $\beta(T)$. Find a basis for the kernel of T .

4. Imagine a static, competitive consumer-choice environment such that a given agent must choose how much of each of n goods to consume, given an income level I in, say, dollars. To fix notation, let $\mathbb{R}_+ = [0, \infty)$, and $X = \mathbb{R}_+^n$ be commodity space, i.e. the set of all bundles of goods available to the consumer. Let $\mathbf{p} \in \mathbb{R}_+^n$ be the vector of prices faced by the agent, i.e. a unit of good i costs p_i dollars.
 - a) As a linear functional, what does $\mathbf{p}'$ measure?
 - b) Write an equation for the agent's budget constraint.
5. Let $T : \mathbf{V} \rightarrow \mathbf{W}$ be linear. Show that the following are equivalent:
 - a) T is invertible;
 - b) $\dim(\mathbf{V}) = \dim(\mathbf{W})$ and $\dim(\ker(T)) = 0$;
 - c) $\dim(\mathbf{V}) = \dim(\mathbf{W}) = \text{rank}(T)$.
6. Let $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\} \subset \mathbb{R}^n$. Let M be the $n \times n$ matrix whose columns are the elements of A . Show that A is a basis for $\mathbb{R}^n$ if and only if $\det(M) \neq 0$.